

SQL Assignment Project: E-Commerce Management System

Project Title

Simple E-Commerce Database System

Objective

Build a simple e-commerce database using SQL. Students will learn:

- Create a database
 - Create tables
 - Use Primary Key & Foreign Key
 - Insert data
 - Perform CRUD operations
 - Write SQL queries using filtering, sorting, joins, and aggregation
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Database Requirements

Create a database named:

ecommerce_db

Tables

1. Customers

Column	Data Type	Constraint
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customer_id	INT	PRIMARY KEY
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name	VARCHAR(100)	NOT NULL
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email	VARCHAR(100)	UNIQUE
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phone	VARCHAR(20)	
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city	VARCHAR(50)	
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2. Categories

Column	Data Type
category_id	INT PRIMARY KEY
category_name	VARCHAR(50)

Example

- Electronics
- Fashion
- Grocery

3. Products

Column	Data Type
product_id	INT PRIMARY KEY
product_name	VARCHAR(100)
price	DECIMAL(10,2)
stock	INT
category_id	INT (Foreign Key)

4. Orders

Column	Data Type
order_id	INT PRIMARY KEY
customer_id	INT (Foreign Key)
order_date	DATE

total_amou DECIMAL(10,2)
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Sample Data Requirement

Students must insert at least

- 5 Customers
 - 5 Categories
 - 10 Products
 - 8 Orders
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Tasks

Part 1 — Database Creation

Create

- Database
 - All tables
 - Primary Keys
 - Foreign Keys
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Part 2 — Insert Data

Insert sample records into every table.

Part 3 — Update

Update

- Product price
- Customer city
- Product stock

Example:

Update Laptop price

Update Customer city

Update Product stock after selling

Part 4 — Delete

Delete

- One customer
 - One product
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Part 5 — Basic Queries

Write SQL queries to:

1.

Show all customers.

2.

Show all products.

3.

Show products whose price is greater than 1000.

4.

Show products whose stock is less than 10.

5.

Show customers from Dhaka.

6.

Sort products by price (Highest to Lowest).

7.

Sort customers alphabetically.

8.

Show the first 5 products.

9.

Count total customers.

10.

Calculate the average product price.

Part 6 — Aggregate Functions

Write queries to find

- Maximum product price
 - Minimum product price
 - Total stock
 - Average stock
 - Total number of orders
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Part 7 — JOIN Queries

1.

Show customer name and their orders.

2.

Show product name with category name.

3.

Show order details with customer name.

Part 8 — Search Queries

Find

- Products containing the word "Phone"
 - Customers whose name starts with "A"
 - Products priced between 500 and 3000
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Part 9 — Bonus Challenge

Write SQL queries to answer:

1. Which product has the highest price?
 2. Which customer placed the largest order?
 3. How many products belong to each category?
 4. Which category has the most products?
 5. List all customers who have placed at least one order.
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Submission Requirements

Students should submit:

- Github link SQL file ([ecommerce.sql](#))
- Screenshots of query outputs
- ER Diagram (optional but recommended)